Shared definitions and lemmas

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Common` (Sections 1–2, Appendix A, Observation 21, Lemmas 19, 22, 29, Eqs. (3), (7))

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

Every result of the formalisation is stated in terms of the definitions in this folder. A wrong definition would make the theorems meaningless, so this overview lists *every* notion of the paper that the Lean statements use, next to its Lean definition, and explains each representation choice. It then lists the shared lemmas and which result folders use them.

1 Conventions

- **Vertex types.** A graph is a Mathlib `SimpleGraph V` on an arbitrary finite type `V` (`[Fintype V] [DecidableEq V]`) with decidable adjacency. Simple, undirected and loopless are built into `SimpleGraph`. The paper's "graphs G, H with vertex sets V, W" are `G : SimpleGraph V`, `H : SimpleGraph W`. Bijections $\pi \in \mathrm{Bij}(V, W)$ are equivalences `V ≃ W`.
- **Matrices** $Q \in \mathbb{R}^{V \times W}$ are functions `V → W → ℝ`; matrices indexed by $V \times W$ (such as B) are `Matrix (V × W) (V × W) ℝ`. All entries are real numbers; the adjacency matrix is real-valued with entries in $\{0, 1\}$.
- **Minima are infima.** Every min in the definitions of $d_\diamondsuit$, $d_\diamondsuit^{\mathrm{OT}}$, $\delta_{p\to q}^{\mathrm{OT}}$ is a Lean infimum (`iInf`). For the finite set $\mathrm{Bij}(V, W)$ this is the minimum. For optimal transport the paper argues that the minima are attained (App. A, pp. 14–15, l. 503–510); this is formalised for the operator norms (`exists_deltaOTPQ_eq`, `exists_dOT_eq`) and the edit distance (`exists_dOTEdit_eq`); for d_1^{OT} and $d_\square^{\mathrm{OT}}$ it is not needed. Infima in $\mathbb{R}$ of empty families are 0: if $|V| \neq |W|$ there is no bijection and no doubly stochastic matrix, and $d_\diamondsuit$, $d_\diamondsuit^{\mathrm{OT}}$ evaluate to 0 (the paper does not define them in this case). Results that need $|V| = |W|$ or $V, W \neq \emptyset$ assume it explicitly.
- **Convention** $0^0 = 0$ (p. 3, l. 84; App. A, p. 14, l. 500) is built into `powZ x a` $= 0$ if $x = 0$, else x^a.
- **Operator norms** are suprema over the unit *ball*; for nonempty index sets this equals the paper's maximum over the unit sphere (App. A, p. 14, l. 479). Lemma 19 is stated with unit balls as well.
- **Normalisation.** For graphs of the same order n, $\delta_\diamondsuit = d_\diamondsuit / s(n)$ and $\delta_\diamondsuit^{\mathrm{OT}} = d_\diamondsuit^{\mathrm{OT}} / s(n)$, with $s(n) = n^2$ for the edit and the cut distance (p. 1, l. 24–27) and $s(n) = n^{\kappa_{p\to q}}$, $\kappa_{p\to q} = 1/q + 1 - 1/p$, for $\|\cdot\|_{p\to q}$ (p. 3, l. 88–90; App. B, p. 15, l. 513–514); so $s(n) = n$ for the p-operator norms $\|\cdot\|_p = \|\cdot\|_{p\to p}$. Lean divides by $n = |V|$ or n^2 (`deltaOne`, `deltaOTOne`, `deltaEdit`, `deltaOTEdit`, `deltaCut`, `deltaOTCut`) and multiplies by $n^{-\kappa}$ in `deltaPQ`; `deltaOTPQ_eq_dOT` proves $\delta_{p\to q}^{\mathrm{OT}} = n^{-\kappa} d_{p\to q}^{\mathrm{OT}}$ from $\mathrm{DS}(V, W) = n\,\mathrm{DS}^*(V, W)$ (App. A, p. 14, l. 494–495).

2 Definitions: paper versus Lean

"(inf)" marks an infimum in place of the paper's minimum.

Paper	Lean	Lean definition / remarks
Graphs and matrices (Sec. 1, p. 1; Sec. 2, p. 5; App. A, p. 14)		
adjacency matrix $A_G \in \{0,1\}^{V\times V}$ (p. 5, l. 181)	`adj G v v'`	1 if `G.Adj v v'`, else 0 (real)
$\deg_G(v) = \lvert N_G(v)\rvert$ (App. A, l. 472)	`deg G v`	number of neighbours (natural number)
$\mathrm{DS}(V, W)$ (p. 5, l. 176–177)	`IsDS Q`	$Q \geq 0$, all row and column sums 1
$P_\pi \in \mathrm{Perm}(V, W)$, $P_\pi(v, w) = 1 \iff \pi(v) = w$ (l. 175–176)	`permMat pi`	as in the paper; `permMat_isDS`

Paper	Lean	Lean definition / remarks
$\operatorname{supp}(Q) = \{(v,w) \mid Q(v,w) > 0\}$ (l. 178)	`Q v w ≠ 0`	for $Q \geq 0$ the same set
A_G^π, i.e. A_G with rows and columns permuted according to π; $A_G^\pi - A_H$ (p. 1, l. 9–10)	`alignedDiff G H pi`	$w, w' \mapsto A_G(\pi^{-1}w, \pi^{-1}w') - A_H(w,w')$, i.e. $A_G^\pi(\pi(v),\pi(v')) = A_G(v,v')$
discrepancy $D_{G,H}(v,w,v',w')$, Eq. (6) (p. 14)	`disc G H v w v' w'`	$A_G(v,v') - A_H(w,w')$
Norms (App. A, Matrix Norms, p. 14)		
ℓ^p norm, $1 \leq p \leq \infty$	`lpNorm p x`	Mathlib's `PiLp` norm; p is an `ENNReal`
$\|A\|_{p\to q} = \max_{\|x\|_p=1} \|Ax\|_q$ (l. 477–479)	`opNorm p q A`	$\sup\{\|Ax\|_q : \|x\|_p \leq 1\}$ (attained: `exists_lpNorm_mulVec_eq_opNorm`)
$\|A\|_{\mathrm{ed}} = \|A\|_{(1)} = \sum \|A(v,w)\|$ (l. 476–477)	`entrywiseNorm`	as in the paper
cut norm $\|A\|_\square = \max_{S,T} \|A(S,T)\|$ (l. 480–481)	`cutNorm A`	sup over pairs of finsets (attained: `exists_eq_cutNorm`)
Hölder conjugate q^* (l. 482–483)	`ENNReal.conjExponent q`	Mathlib; $1/q^* = 1 - 1/q$ (`inv_conjExponent_toReal`)
Costs and local costs (Sec. 2, p. 5; App. A, p. 14)		
$L_Q(v,w) = \sum_{(v',w')} \|D(v,w,v',w')\|\, Q(v',w')$ (l. 187–188)	`localCost G H Q v w`	as in the paper
$C_1(Q) = \|B_{1\to1}(G,H,Q)\|_1 = \max_{\mathrm{supp}(Q)} L_Q$ (l. 185, 188; Eq. (7))	`costOne G H Q`	*defined* as $\sup_{(v,w)\in\mathrm{supp}(Q)} L_Q(v,w)$; equality with $\|B_{1\to1}\|_{1\to1}$ is proved (`opNorm_Bpq_one_one_eq_costOne`)
$C_{\mathrm{ed}}(Q) = \|B(G,H,Q)\|_{\mathrm{ed}}$ (l. 184–185)	`costEdit G H Q`	$\sum \|D\|\, Q(v,w)Q(v',w')$; $= \|B\|_{\mathrm{ed}}$ for $Q \geq 0$ (`costEdit_eq_entrywiseNorm_B`)
$C_\square(Q) = \|B(G,H,Q)\|_\square$ (l. 184–185)	`costCut G H Q`	`cutNorm (B G H Q)`
$C_\diamond(\pi) = C_\diamond(P_\pi) = \|A_G^\pi - A_H\|_\diamond$ (l. 185–186)	`costOneBij`, `costEditBij`	$C_\diamond(P_\pi)$; the identity with $\|A_G^\pi - A_H\|$ is `costEditBij_eq`, `localCost_permMat_apply`
Fractional alignments and B-matrices (Sec. 1, pp. 2–3)		
$\mathrm{DS}^*(V,W)$: $Q^* \geq 0$, row sums $1/\|V\|$, column sums $1/\|W\|$ (p. 2, l. 50–54; App. A, l. 493–494)	`IsCoupling Q`	as in the paper
$B(G,H,Q)((v,w),(v',w')) = Q(v,w)\, D\, Q(v',w')$ (p. 2, l. 68–69)	`B G H Q`	as in the paper
$B_{p\to q}(G,H,Q) = Q(v,w)^{1/q}\, D\, Q(v',w')^{1-1/p}$ (p. 3, l. 84–85)	`Bmat p q G H Q`	with `powZ`, exponents $1/q$, $1-1/p$ (so $1/\infty = 0$)
$\kappa_{p\to q} = 1/q + 1 - 1/p$ (App. B, p. 15, l. 513)	`kappa p q`	as in the paper
Distances		
$d_\diamond(G,H) = \min_\pi \|A_G^\pi - A_H\|_\diamond$ (p. 1, l. 7–9)	`dOne`, `dEdit`, `dCut`, `distPQ p q`	(inf) over `V ≃ W`
$d_\diamond^{\mathrm{OT}} = \min_{Q\in\mathrm{DS}(V,W)} C_\diamond(Q)$ (p. 5, l. 184–187)	`dOTOne`, `dOTEdit`, `dOTCut`, `dOT p q`	(inf) over `IsDS`; graphs of the same order
$\delta_\diamond = d_\diamond/s(n)$, $\delta_\diamond^{\mathrm{OT}} = d_\diamond^{\mathrm{OT}}/s(n)$ with $s(n) = n^2$ for ed, $\square$ (p. 1, l. 24–27) and $s(n) = n^{\kappa_{p\to q}}$ for $p \to q$ (p. 3, l. 88–90; App. B, l. 513–514)	`deltaOne`, `deltaOTOne`, `deltaEdit`, `deltaOTEdit`, `deltaCut`, `deltaOTCut`, `deltaPQ`	division by $n = \|V\|$ resp. n^2; `deltaPQ` multiplies by $n^{-\kappa}$

Paper	Lean	Lean definition / remarks
$\delta^{\mathrm{OT}}_{p\to q} = \min_{Q^*\in \mathrm{DS}^*} \|B_{p\to q}(G,H,Q^*)\|_{p\to q}$, Eq. (2) (p. 3, l. 87)	`deltaOTPQ p q G H`	(inf; attained, `exists_deltaOTPQ_eq`) over couplings, any orders; $= n^{-\kappa} d^{\mathrm{OT}}_{p\to q}$ for equal orders (`deltaOTPQ_eq_dOT`)
$\delta^{\mathrm{OT}}_{\mathrm{ed}}$, Eq. (1), and $\delta^{\mathrm{OT}}_{\square}$ via couplings (p. 2, l. 66, 77)	`deltaOTEdit`, `deltaOTCut`	only for graphs of the same order, via $\mathrm{DS}(V,W)$ and n^{-2} (the coupling form for different orders is not used by any formalised result)
blowup $G^{\bullet k}$, $\delta^{\bullet} = \lim_\ell \delta(G^{\bullet n\ell}, H^{\bullet m\ell})$ (p. 1, l. 36–42)	`blowup`, `deltaBlowup`	in `Definitions.lean` of `Theorem1_BlowupEqualsOT`; vertex set $V\times[k]$, $(v,i)\sim(v',i') \iff v\sim v'$; `limUnder`
$\zeta_Q(A,B) = \|D_{A,B}\|_{\square,\mu_Q}$, $\mu_Q = Q/n$ (App. E.2.2, p. 36, l. 1129–1139; Eq. (21), p. 32)	`epsR A B Q`	cut norm of $\mu_Q\otimes\mu_Q\cdot D_{A,B}$; $= n^{-2}C_{\square}(Q)$ for graphs (`epsR_adjMat`)
$\rho_H(w,w') = \|N_H(w)\triangle N_H(w')\|$ (p. 28, l. 866)	`rho H w w'`	$\sum_{w''} \|A_H(w,w'') - A_H(w',w'')\|$

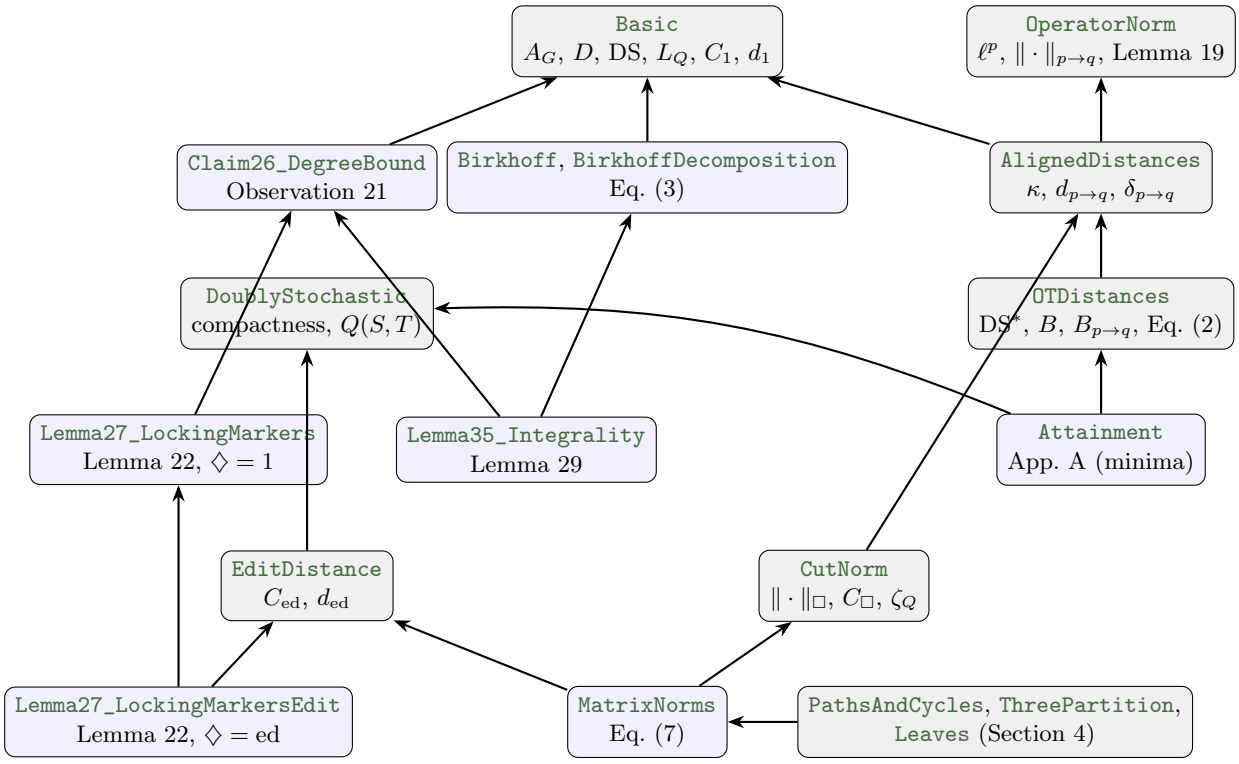
3 Shared lemmas and where they are used

Paper	Lean (file)	Used by (folder: result)
Lemma 19 (p. 14)	`lemma24`, `abs_dotProduct_mulVec_le_opNorm`, `exists_abs_dotProduct_mulVec_eq_opNorm` (`OperatorNorm.lean`)	`Theorem1`: Thm. 1; `Proposition14`: Prop. 13; `Theorem15`: Thm. 14
Eq. (3), Birkhoff (p. 5)	`exists_bij_subset_supp` (`Birkhoff.lean`)	`Theorem10`: Thm. 9 (reference bijection, Eq. (18)); inside Common: Lemma 29
	`exists_birkhoff_decomposition`, `exists_birkhoff_decomposition_pos`, `isDS_of_decomposition` (`BirkhoffDecomposition.lean`)	`Theorem7`: Thm. 6; `Theorem15`: Thm. 14
Eq. (7) (p. 14)	`opNorm_Bpq_one_one_eq_costOne`, `dOT_one_one_eq_dOTOne` (`MatrixNorms.lean`)	`Theorem17`: Thm. 16
App. A, minima attained	`exists_deltaOTPQ_eq`, `exists_dOT_eq` (`Attainment.lean`)	`Theorem1`: Claim 20; `Theorem15`: Thm. 14
	`exists_dOTEdit_eq` (`EditDistance.lean`)	`Theorem16`: Thm. 15
normalisation, $\delta^{\mathrm{OT}}_{p\to q} = n^{-\kappa} d^{\mathrm{OT}}_{p\to q}$ (Secs. 1–2)	`deltaOTPQ_eq_dOT` (`OTDistances.lean`)	`Proposition14`: Prop. 13; `Theorem15`: Thm. 14
Observation 21 (p. 20)	`claim26`, `claim26_bij` (`Claim26_DegreeBound.lean`)	inside Common: Lemma 22 (1) and Lemma 29; hence Props. 5, 8, Thm. 9
Lemma 22, $\diamondsuit = 1$ (pp. 20–21)	`lockGraph`, `lemma27_bij`, `lemma27_cost` (`Lemma27_LockingMarkers.lean`)	`Proposition9`: Prop. 8; the construction `lockGraph` also `Proposition6`: Prop. 5
Lemma 22, $\diamondsuit = \mathrm{ed}$	`lemma27_bij_edit`, `lemma27_cost_edit` (`Lemma27_LockingMarkersEdit.lean`)	`Proposition6`: Prop. 5
Lemma 29 (p. 28)	`lemma35_rho_le`, `lemma35_iso`, `lemma35_dOne_eq_zero`, `lemma35_one_le_dOTOne` (`Lemma35_Integrality.lean`)	`Proposition9`: Prop. 8 (Lemma 27); `Theorem10`: Thm. 9

Paper	Lean (file)	Used by (folder: result)
cut norm, $C_\square$, ζ_Q	`CutNorm.lean`	`Proposition11`: Prop. 12; `Proposition12`: Prop. 10; `Theorem13`: Thm. 11; `Theorem22`: Thm. 18
C_{ed}, d_{ed}, $d_{\mathrm{ed}}^{\mathrm{OT}}$	`EditDistance.lean`	`Proposition6`: Prop. 5; `Theorem7`: Thm. 6; `Theorem16`: Thm. 15
paths, cycles, Three-Partition (App. G)	`PathsAndCycles.lean`, `ThreePartition.lean`	`Theorem16`–`Theorem18`: Thms. 15–17
attached leaves (App. G)	`Leaves.lean`	`Theorem18`: Thm. 17; `Theorem22`: Thm. 18

(Folder names are abbreviated: `Theorem10` is `Theorem10_LocalEditUpperBound`, etc.)

4 Module structure



Arrows point from a module to (some of) the modules it imports; `DoublyStochastic` imports `Basic`, `MatrixNorms` imports `OTDistances`. Grey: definitions; white: shared lemmas.

5 The shared lemmas: paper versus Lean

- **Lemma 19** (p. 14, l. 488–492; $\|A\|_{p\to q} = \max_{\|y\|_{q^*}=1,\|x\|_p=1} |y^\top Ax|$): stated with unit balls; proved via Hölder (`abs_dotProduct_le_lpNorm_mul`) and its equality case (`exists_dotProduct_eq_lpNorm`), as in the paper.
- **Eq. (3)** (p. 5, l. 193–196): `exists_birkhoff_decomposition_pos` is the exact statement ($\lambda_i > 0$, $\sum \lambda_i = 1$). `exists_birkhoff_decomposition` indexes by all permutations with $\lambda \geq 0$, which Theorems 6 and 14 use. All are derived from Mathlib's Birkhoff–von Neumann theorem.
- **Eq. (7)**: $\|B_{1\to 1}(G,H,Q)\|_{1\to 1} = \max_{\mathrm{supp}(Q)} L_Q$ for $Q \in \mathrm{DS}(V,W)$, the identity $C_1(Q) = \max_{(v,w)\in\mathrm{supp}(Q)} L_Q(v,w)$ of Section 2 (p. 5, l. 188–189). Eq. (7) (p. 14, l. 500–502) is the same identity for couplings $Q^* \in \mathrm{DS}^*(V,W)$, inside the minimum defining δ_1^{OT}.

- **App. A, minima attained** (pp. 14–15, l. 503–510): the paper's argument (compactness; for operator norms, restricting $B_{p\to q}(G, H, Q_k)$ to the rows and columns indexed by $\operatorname{supp}(\lim Q_k)$ gives convergent matrices and does not increase the norm) is `opNorm_restrict_le` and `lowerSemicontinuous_opNorm_Bmat`.
- **Observation 21** (p. 20, l. 642–643; $L_Q(v, w) \geq |\deg_G(v) - \deg_H(w)|$): triangle inequality and the row/column sums of Q, as in the paper.
- **Lemma 22** (locking markers, pp. 20–21, l. 650–674): new vertex set $V \oplus [m]$, clique vertex y_{i+1} = `Sum.inr i`, marker index j = `j.val + 1`; Δ, τ, δ are natural numbers (the paper: integers $\tau \geq 1$, $\delta \geq 0$). Both items for $\diamondsuit = 1$ follow the paper; for $\diamondsuit =$ ed, item 1 is derived from $\diamondsuit = 1$ via $C_1(\pi) \leq C_{\mathrm{ed}}(\pi)$ and item 2 is proved directly.
- **Lemma 29** (p. 28, l. 872–886): stated for any $Q \in \mathrm{DS}(V, W)$ whose supported local costs are at most L (the paper: an optimal Q and $L = d_1^{\mathrm{OT}}$), with a bijection σ inside $\operatorname{supp}(Q)$ (Eq. (18)).

6 How to check

```
lake build GsimOT.Common.MatrixNorms GsimOT.Common.Attainment GsimOT.Common.Lemma27_LockingMarkersEdit
           GsimOT.Common.Lemma35_Integrality GsimOT.Common.BirkhoffDecomposition GsimOT.Common.Leaves
```

or `bash build.sh` for everything. The table of all correspondences is `Common_correspondence.pdf`.